Contrastive CUR: Interpretable Joint Feature and Sample Selection for Case-Control Studies

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1 ENRICHMENT PATHWAYS

1.1 Mouse Protein

Method	# Enriched Pathways (FDR < 0.05)	# Multi-Gene Enriched Pathways (FDR < 0.05)
CCUR	440	83
CUR	282	51
CPCA	361	100
Multi-CPCA	407	83
CFS	334	70
Welch t-test	476	88
Wilcoxon	410	74

Table 1. Mouse Protein Pathway enrichment comparison across methods. We report the number of enriched pathways at a false discovery rate (FDR) threshold of 0.05, as well as the number of enriched pathways supported by at least two selected genes.

1.2 Small Molecule

Method	# Enriched Pathways (FDR < 0.05)	# Multi-Gene Enriched Pathways (FDR < 0.05)
CCUR	132	7
CUR	95	24
CPCA	120	57
Multi-CPCA	84	42
CFS	92	23
Welch t-test	34	27
Wilcoxon	41	32

Table 2. Small Molecule Pathway enrichment comparison across methods. We report the number of enriched pathways at a false discovery rate (FDR) threshold of 0.05, as well as the number of enriched pathways supported by at least two selected genes.

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1.3 Pathogen

Method	# Enriched Pathways (FDR < 0.05)	# Multi-Gene Enriched Pathways (FDR < 0.05)
CCUR	53	2
CUR	15	0
CPCA	150	60
Multi-CPCA	80	8
CFS	61	7
Welch t-test	71	7
Wilcoxon	72	5

Table 3. Pathogen Pathway enrichment comparison across methods. We report the number of enriched pathways at a false discovery rate (FDR) threshold of 0.05, as well as the number of enriched pathways supported by at least two selected genes.